MODULE 2
Practical 1 : Introduction to NumPy Arrays

NumPy (Numerical Python) is a Python library used for working with arrays and performing mathematical operations efficiently.

Main Uses of NumPy Arrays

1. **Store numerical data** (integers, floats, etc.).
2. **Perform mathematical operations** such as addition, subtraction, multiplication, and division.
3. **Matrix operations** like matrix addition, multiplication, and transpose.
4. **Statistical calculations** such as mean, median, standard deviation, maximum, and minimum.
5. **Data analysis and machine learning** applications.
6. **Scientific and engineering calculations**.
7. **Easy indexing and slicing** of data.
8. **Multi-dimensional arrays** (1D, 2D, 3D, etc.).

How to install numpy

1) win+R

2)cmd

3) type py –m pip install numpy

• Write a Python program to import NumPy library.

```
import numpy as np

# Create a one-dimensional array
arr_1d = np.array([10, 20, 30, 40, 50])

print("1-D Array:", arr_1d)
print("Shape:", arr_1d.shape)
print("Dimensions:", arr_1d.ndim)
print("Data Type:", arr_1d.dtype)
```

Create two-dimensional NumPy arrays.

```
import numpy as np

# Create a two-dimensional array

arr_2d = np.array([[1, 2, 3],

                   [4, 5, 6],

                   [7, 8, 9]])

print("2-D Array:")

print(arr_2d)

print("Shape:", arr_2d.shape)

print("Dimensions:", arr_2d.ndim)

print("Total Elements:", arr_2d.size)
```